Sony SA3800 Standard Operating Procedure & User Guide
UC Davis Flow Cytometry Shared Resource

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Computer Login Details: **Account**: FCM **Password**: FCM

Part 1: System Startup

1. Turn on the SA3800. (Power button on front)
2. Check DI Water level (Sheath).
    a. If the DI Water tank is not full:
        i. Remove the lid from the DI Water tank and place into the sheath lid holder to keep the line clean.
        ii. Fill the DI Water tank with MilliQ water.
            1. Take a right outside of SA3800 room, then a right at the first hall, sink on your right has MilliQ System.
            2. Press the "Non-Stop" button on the MilliQ system control panel.
            3. Press the dispense button on the top of the MilliQ spout to test for water flow then press again to disengage.
            4. Adjust the MilliQ spout over the tank opening.
            5. Press the dispense button again and fill the tank.
        iii. Return the Water tank to the SA3800 and reattach the lid.
        iv. Make a note to email JVD as the Water tank should have been filled at the end of every run if <50% full.
    b. If the DI Water tank is full, move to the next step.
3. Check Waste tank level.
    a. If the Waste tank is >50% full:
        i. Remove the lid from the Waste tank and place it into the waste lid holder.
        ii. Take the Waste tank to the sink and dump it down the drain.
        iii. Refill the Waste tank with 1L of 100% bleach (under the sink).
        iv. Return the Waste tank to the SA3800 and reattach the lid.
        v. Make a note to email JVD as the Waste tank should have been emptied at the end of every run if >50% full.
    b. If Waste level is <50% move to the next step *If you have a large experiment, you should start with a freshly prepared Waste tank by following step 3a.
4. Open the SA3800 software.
5. Log into the SA3800 software
    a. Account name: "PI last name (capital first letter)" + "Lab". E.g. SmithLab
    b. Password: "PI Last name" + "Current year or previous year". E.g Smith2024
    c. Note: The SA3800 may force users to change their password every year (365days). When this happens, please update the password on your lab's account to "PI last name (capital first letter)" + "Current year".
6. Allow the system to connect.

7. Remove the tube rack from the inside system.
8. Wait for System Prime window to popup.
9. Click *Start* in the System Prime window.
    a. This will take about 5min to complete.

Part 2: Optical Alignment and Performance QC

1. Obtain *Sony Beads Control* box from the fridge.
2. Obtain the date and time of the last QC run:
    a. Click the *QC* tab on the left side of the screen.
    b. Ensure the "Previous Run" radio button in the *View* box is selected.
    c. Note the last time the Daily QC was run and its result.
3. If you are the first user of the week or the last QC was completed >5 days prior perform the following clean:
    a. Contrad cleaning procedure:
        i. Click the *Cytometer* tab at the top.
        ii. Click the "Decontamination" button.
        iii. Select the "Bleach Cleaning and Rinse" from the drop down.
        iv. Follow the cleaning wizard with the following adjustments:
            1. Bleach tube: Use a **20% Contrad** solution instead
                a. Fill fresh FACS tube to weld line with 100% contrad then fill ⅔ with DI water.
            2. Rinse tube: Use 100% Di Water
4. If you are the first user of the week perform the following Optical Alignment:
    a. Optical Alignment:
        i. Click the *Cytometer* tab at the top.
        ii. Click the "Optics Alignment" button.
        iii. Follow the "Optics Alignment" wizard.
            1. Procedure uses **AlignCheck** beads.
            2. To prepare the **AlignCheck** bead control (do this every time):
                a. Carefully add ~450uL of fresh DI water into a fresh tube.
                b. Vortex **AlignCheck** bead stock right-side up, upside down and sideways.
                c. Carefully add 2 drops of the AlignCheck beads to the tube.
                d. Label and Vortex the tube.
            3. Make sure to check the bead lot of the AlignCheck beads with the drop down in wizard.
5. If the last QC run date is from the current day skip to step 6, otherwise run Daily QC. ***Note: You should always run Daily QC if you ran the above Optical Alignment.***
    a. Click the *QC* tab on the left side of the screen.
    b. Click the "Align Check QC" button.
    c. Click "Start Daily QC", **follow the wizard with the notes below**:
        i. Procedure uses **AlignCheck** beads.
        ii. To prepare the **AlignCheck** bead control (do this every time):
            1. Carefully Carefully add ~450uL of fresh DI water into a fresh tube.
            2. Vortex **AlignCheck** bead stock right-side up, upside down and sideways.
            3. Carefully add 2 drops of the AlignCheck beads to the tube.

4. Label and Vortex the tube.

iii. Make sure to check the bead lot of the AlignCheck beads with the drop down in wizard.

d. Click the latest Daily QC run and confirm the result.

1. If the QC passed, move onto the next step.
2. If the QC fails or has a warning, identify the warning, notify Flow Core Personnel, and run a Contrad cleaning procedure for the system.
    a. Contrad cleaning procedure:
        i. Click the *Cytometer* tab at the top.
        ii. Click the "Decontamination" button.
        iii. Select the "Bleach Cleaning and Rinse" from the drop down.
        iv. Follow the cleaning wizard with the following adjustments:
            1. Bleach tube: Use a **20% Contrad** solution instead
                a. Fill fresh FACS tube to weld line with 100% contrad then fill ⅔ with DI water.
            2. Rinse tube: Use 100% Di Water

6. The system is QCed and ready to run.

Part 3: Compensation/unmixing using the Reference Library (you will setup a separate compensation experiment)

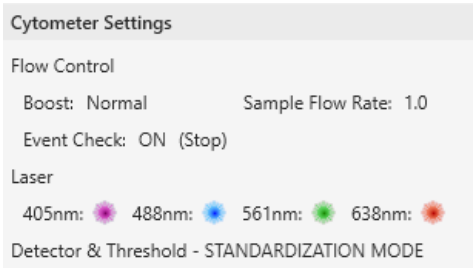
- Compensation controls need to be run at least once for each fluorochrome you plan to use.
- Tandem dye fluorochrome compensation controls need to be run for **each new product or lot of the same product.**
- Tandem dye fluorochrome products should have their **reference spectra updated/refreshed every month**.
- Good quality spectra that almost saturate (>10^5) the primary detector for a given fluorochrome will give the best results (adjusted with detector array gain).
- Bring a printout with details about each fluorescent reagent in your panel:
    - Fluorochrome name
    - Product lot
    - Antibody clone (If applicable)
    - Relative intensity of fluorochrome,
        - Flurofinder is a great place to find this information.

Setting up the compensation experiment

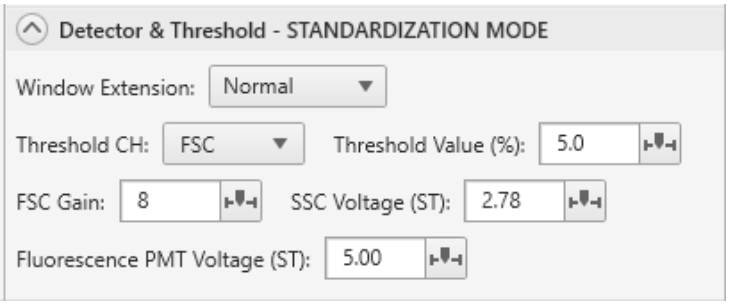
1. Click on the *Preparation* tab.
2. Click on the "Spectral Library" button.
3. Using the *Add New Spectral Reference* box:
    a. Type the name of each fluorochrome in your assay (you will need a single-color control for each marker in your assay), select it from the filtered list, click "Add New".
    b. If your fluorochrome is not in the list:
        i. Add a custom fluorochrome by fully typing the name of your Fluorochrome into the box and then click "Add New".

        ii. In the “Add New Flurochrome” popup window, set the *Start* detector to the detector that captures the peak emission of the fluorochrome, then set the *End* detector to Start + 1.
4. CRITICAL: For each fluorochrome, click the “Laser” box and select “405/488/561/638”. Note: This ensures that your spectral references are taken with all lasers active. Failure to select all lasers will severely limit how you can use the newly generated spectral references.
5. Add reagent and marker information:
    a. Label: This should be something to help differentiate each set of markers (IE panel name)
    b. Marker name (IE CD3)
    c. Species
    d. Antibody clone (If applicable)
    e. Manufacturer
    f. Product lot
6. Once all fluorochromes have been added click the “Single Stain Control” button from the bottom.
7. Choose the plate type.
    a. 24-well tube rack/96-well plate (std/half deep/deep)/384 well-plate
8. Confirm the tube rack/plate layout matches your desired layout.
    a. Tube/Well coordinates can be customized by typing in an alternative tube/well coordinate into the position text box for each sample.
9. The “Use a common unstained control for single stain controls” box:
    a. If you have a MATCHED internal unstained population within each of your single color controls then uncheck the “Use a common unstained control for single stain controls” box.
        i. This will ensure you will have both positive and negative gates automatically created for each single color control. (With the box checked each single color control only have a positive gate)
    b. If you do NOT have a MATCHED internal unstained population within each of your single color controls then keep the “Use a common unstained control for single stain controls” box checked.
        i. This will ensure you will have ONLY a positive gate created for each single color control and a single unstained sample will be used for all single color control negatives.
    c. Note: There is some experiment specific nuance for this setting, your trainer will go over any deviation from the above.
10. Click “Next”
11. Set the *Acquisition Settings*
    a. Select the order for acquisition to your preference
    b. Cleaning Speed: Normal
    c. Acquisition Offset Time: 4sec
    d. Select “Enable sample agitation”. *Note: This shakes the plate and adds time to run.
        i. Set period and duration according to your preference.
            1. Recommended: Agitate every 1 wells
12. Click “Next”
13. Select “Histrogram” as the Worksheet plot type for gating the single color controls.
14. Click “Next”
15. Enter a name for your experiment. (Recommended Name: *Panel Name – Compensation*)
16. Click “Next”

17. Review experiment details and layout.
    a. If you are using tubes, vortex and load each single-color control into its proper position in the tube rack.
    b. If you are using a plate, ensure the sample (software) layout matches your plate layout, if it does not customize the sample (software) layout to match your plate.
18. CRITICAL: In the *Cytometer Settings* box confirm *Detector and Threshold - STANDARDIZATION MODE* is listed as system mode. If it is not, please reach out to flow core staff for assistance.
19. Click “Next”



Acquire and Save Single Color Controls to Reference Library



20. Set initial values for Threshold, FSC, SSC and Detector gain in the *Detector and Threshold* box **Note these are only suggested initial settings you may need to customize for your single color controls.*
    a. Set each of the follow to the stated value:
        i. Window Extension: “Normal”
        ii. Threshold channel: “FSC”
        iii. Threshold Value (%): “5.0”
        iv. FSC Gain: “8”
        v. SSC voltage: “2.78”
        vi. Fluorescence PMT Voltage: “5.00”
21. Follow the on screen Single Stain Controls Guide.
    a. In this section you will need to identify the brightest fluorochromes in your panel. Fluorochromes that have the same or an unknown level of intensity will all need to be run to determine who is the brightest.
        i. Fluorofinder is a great free resource to identify fluorochrome intensity and peak emission. (https://fluorofinder.com/)
    b. Common brightest fluorochromes include:
        i. PE > BV421 > APC >AF647

Part 4: Setting up and Running the Experiment

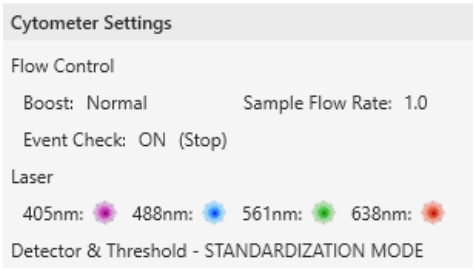
- This SOP is expecting that you will have all reference controls already in the Reference Library, if you have new colors or need to refresh a 1 or more controls in the Reference Library, please follow the previous section with just those markers and then return to this point.

Setting up the acquisition experiment

1. Click on the *Preparation* tab.
2. Click on the “Spectral Library” button.

3. Add each fluorochrome's reference spectra from the Reference Library (top section of window) in your assay to the *Selected Spectral References* box:
    a. "CTRL" clicks each reference spectra for each fluorochrome (<u>you must have a reference for each fluorochrome in your assay</u>) in your assay.
    b. Click the "Select" button to add the selected reference spectrums to your experiment.
4. Click on the "Experiment Designer" button.
5. Choose the rack/plate type.
    a. 24-well tube rack/96-well plate (std/half deep/deep)/384 well-plate
6. Click "Next"

7. Set the *Acquisition Settings*
    a. Select the order for acquisition to your preference
    b. Cleaning Speed: Normal
    c. Acquisition Offset Time: 4sec
    d. Select "Enable sample agitation". *Note: This shakes the plate and adds time to run.
        i. Set period and duration according to your preference.
            1. Recommended: Agitate every 1 wells



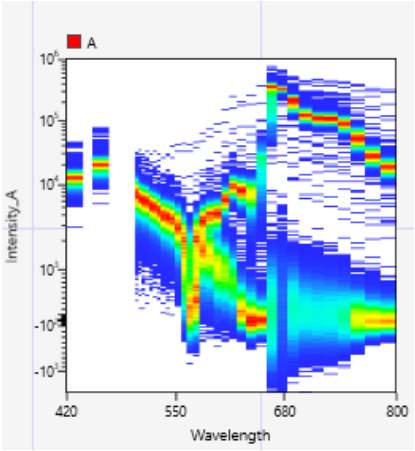
8. Click "Next"
9. Enter a name for your experiment. (Recommended Name: *Panel Name – Date*)
10. Click "Next"
11. Select "Standardization Mode (beta)" radio button.
12. Click "Next"
13. Review experiment details.
14. CRITICAL: In the *Cytometer Settings* box confirm *Detector and Threshold - STANDARDIZATION MODE* is listed as system mode. If it is not, press the "Back" button then follow Step 11.
15. Click "Create Experiment"
16. Click on the text (not check box) of the name of the first tube rack/plate in the Experiment Layout.
17. Click tube/well "A1" from the tube rack/plate map.
18. Click on the FSCXSSC plot.
19. Click the *Y-Axis* drop down (top of screen) and select "Log".
20. Label tubes/wells in software:
    a. Click the "Map" button, upper right side of plate map. 
    b. Edit plate map to label each well.
    c. Multi-select each sample tube/well
    d. In the *Sample Settings* box:
        i. Use the drop down to add samples to the desired group.
        ii. Click the "Auto Acquisition Target" radio button *Note: Skip if you plan to run each manually.
    e. If using the tube rack, load your tubes into the rack according to the map.

Setting the gain/voltage for FSC, SSC and detector array to maximize sensitivity.

- In this next section you will use the Unstained sample to set your FSC and SSC gains and then use your brightest full stained sample to adjust the gains for the detector array. The goal is to use as high a gain as possible but still ensuring no detector within the array is off scale. Pay extra attention to the detector channels corresponding to your brightest fluorochromes.

21. Set initial values for Threshold, FSC, SSC and Detector gain in the *Detector and Threshold* box **Note these are only suggested initial settings you may need to customize for your single color controls.*
    a. Set each of the follow to the stated value:
        i. Window Extension: “Normal”
        ii. Threshold channel: “FSC”
        iii. Threshold Value (%): “5.0”
        iv. FSC Gain: “8”
        v. SSC voltage: “2.78”
        vi. Fluorescence PMT Voltage: “5.00”
22. Load the sample rack or plate into the SA3800.
23. Click the “Load Tube Rack/Plate” button. 

24. Highlight the tube/well with the “Unstained” control.
25. Right click the tube/well and select “Set Current Position”.
26. In the *Flow Contro*l box set the Sample Flow Rate: “1”
27. Click the “Preview” button to begin acquiring the sample.
28. While sample is running:
    a. Monitor the In the FCSxSSC Plot and using the *Detector and Threshold* box adjust the FSC gain and SSC voltage to bring your cellular population/s on scale and above the threshold.
29. Click the “Stop” button to stop sample acquisition.
30. Highlight the tube/well with the brightest “Full Stained” sample.
31. Right click the tube/well and select “Set Current Position”.
32. Click the “Preview” button to begin acquiring the sample.
33. In the Spectral Ribbon plot:
    a. Using the *Detector and Threshold* box, adjust the gain for the detector array to ensure the brightest events are at 10^5 or above but remain on scale (<10^6). **See example below*
        i. Generally, you will want to use a gain between 4.5-5.5
        ii. Note: None of your **<u>singlet events</u>** should be above 10^6 on ANY detector.



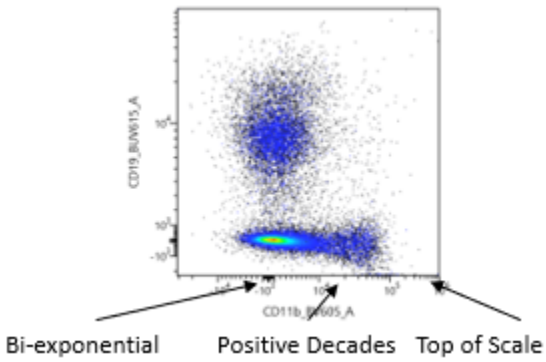
Recording the experimental samples

22. In the *Stopping Condition* box, set the events to record, be careful not to run the tube/well dry.
    a. Recommended: 10,000+ depending on population rarity. *For rare <1% populations you may need to record 100,000 or more events!*

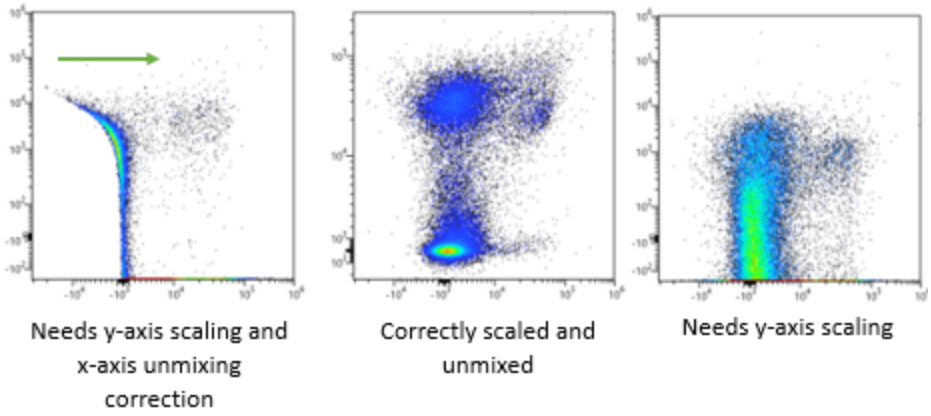
23. Click “Auto Acquire” button  to begin recording your tube rack/plate.
24. Click the “Unload” button to unload the tube rack/plate.

Unmix the data and make adjustments to the unmixing

25. Select a full stained sample from the rack/plate map.
26. Create the plots and gates needed to remove debris and identify viable single cells.
    a. FSC x SSC (morphology), SSC-H x SSC-W (Singlets), Viability Dye x SSC-A (Viable Cells)
27. Right click the tube/well and select “Copy”.
28. Highlight all other tubes/wells (that you would like to have the same plots) and right click and high light paste and select “Worksheet”.
29. Select a full stained sample from the rack/plate map once again.
30. Select the *Unmixing* tab at the top.
31. In the *Unmixing* box
    a. From the *Algorithm* drop down, select "Spectrum"
    b. Click the “Apply” button *Note: WLSM should be in the upper right corner of each plot.
32. Click the “Spectral Reference Adjuster” button in the *Matrix* box at the top and from the drop down select the single cells population you created above. Note: Make sure to include the dead cells to ensure the viability dye was unmixed correctly.
33. Fix the scaling and any unmixing errors by doing the following for each parameter in your assay:



    a. ***Negative decades***, ***width basis/positive decades,*** and ***top of scale*** for each plot to fix visual clustering anomaly around zero, properly visualize events below/around the axis and eliminate white space in the top end . (see image above for scaling adjustment anchors)
    b. For each plot click and drag over/under compensated populations to adjust the unmixing.
        i. This can be very touchy, make small ½ pixel adjustments first.
        ii. Populations should appear rounded and generally straight. If a diagonal is witnessed, adjustment may be necessary (this means there is still correlation between the two markers in that particular plot, this could be an unmixing error or a biological phenomenon; talk to flow core personnel to discuss the particulars of your assay).



34. After all parameters have been checked, if adjustments were made click the “OK” button, otherwise press “Cancel”.
35. If necessary, repeat the above “Spectral Reference Adjuster” steps with just the viable cells to better visualize your healthy cells. This can often make correcting Unmixing easier to visualize.

Export FCS Files

- **It is recommended that you export your FCS unmixed (compensated) instead of raw. This is especially critical if you used controls from your Reference Library. If you instead ran a fresh set of compensation controls the day of your experiment you can export the data in a raw format (much larger file sizes), making sure to include the raw compensation data files, and unmix in FlowJo. If you opt for this approach, please speak to core personnel about the procedure to unmix in FlowJo.**

36. Right click on the “Sample Group” you want to export and select “Export to FCS File” in the menu.
37. In the *FCS File Export* window:
    a. Check the checkbox for all sample groups you wish to export.
    b. Select the Unmixing matrix you want to use from the *Matrix* box.
    c. Select “WLSM” from the *Algorithm* box
    d. Select 3.1 from the “FCS version” drop down
    e. Click the “Browse” button to direct your FCS file export to your labs export folder (C:\Users\fcm\desktop\FSC Export***Your PI’s last name + Lab***
    f. Click the “Export” button.
    g. Once the progress bar is finished your export will begin: (need to check if SA3800 works same as ID7000 here)
        i. *The actual export occurs OUTSIDE the SonySA3800 software and can take awhile. To track the progress please check in the bottom right side of the “Task Bar/Start Menu” near the clock and click “^”, then mouse over the FCS Export applet.*

Part 5: System Clean and Shutdown

System Clean

- **Complete this cleaning after each experiment.**

38. Click the *Cytometer* tab at the top.
39. Click the “Decontamination” button.
40. Select the “Bleach Cleaning and Rinse” from the drop down.

41. Follow the Bleach Cleaning and Rinse wizard.
    a. Use a 10% bleach for cleaning solution
    b. Use DI water (squirt bottle) for rinse solution

System Shutdown

- **Only run this if you are the last person of the day.**

42. Click the *Cytometer* tab at the top.
43. Click the “Hardware Shutdown” button.
44. Select the “Bleach Cleaning and Rinse” from the drop down.
45. Follow the Hardware Shutdown wizard.
    a. Use a 10% bleach for cleaning solution
    b. Use DI water (squirt bottle) for rinse solution
46. Click the “Shutdown” button when the wizard completes.